Solving the corresponding difference equation

Last time we have concluded that the solution to the finite horizon optimal growth model is equivalent to the solution of:

$$(**)\quad \boxed{\begin{array}{l} u'(f(k_t)-k_{t+1}) = \beta u'(f(k_{t+1})-k_{t+2})\cdot f'(k_{t+1}) \qquad t=0..T-1 \\ k_0 \text{ is given} \\ k_{T+1}=0 \quad (\text{by property } 1) \end{array}}$$

In general, second-order difference equations are hard to solve.
It can be done numerically or, for some $u(\cdot)$ & $f(\cdot)$, analytically.

1 Numerical solution – "shooting" algorythm:

Given k_0

loop
- Guess k_1
  - find $k_2 \Rightarrow$ find $k_3 \Rightarrow ... \Rightarrow$ find $k_T \Rightarrow$ find k_{T+1}
- If $k_{T+1}=0 \Rightarrow$ done
- If $k_{T+1}\neq 0 \Rightarrow$ adjust k_1

Remark: for k_t, k_{t+1} : k_{t+2} solves

$$\underbrace{\frac{u'(f(k_t)-k_{t+1})}{\beta f'(k_{t+1})}}_{\text{LHS, indep. of } k_{t+2}} = \underbrace{u'(f(k_{t+1})-\underbrace{k_{t+2}}_{\text{unknown}})}_{\text{RHS, }\uparrow \text{ in } k_{t+2}}$$

if $RHS(k_{t+2}=0) > LHS \Rightarrow \nexists\ k_{t+2}$ solving this equation $\Rightarrow$ adjust the guess on k_1 (the solution must exist if optimal k_1 is chosen)

Note also that if $f(0)>0 \Rightarrow$ corner solution $k_{t+1}=0$ is possible, then EE may hold with inequality, and the solution (numerical algorythm) should account for it.

② Analytical solution can be found for some $u(\cdot)$, $f(\cdot)$ namely, $u(c)=\log c$ or $u(c)=\frac{c^{1-\delta}-1}{1-\delta}$

$f(k)=Rk$ or $f(k)=Ak^{\alpha}$

Example 1: $u(c)=\log c$, $f(k)=Rk$

For linear production technology the solution is simple

$$\begin{cases} c_0+k_1=Rk_0 \\ \frac{c_1}{R}+\frac{k_2}{R}=k_1 \\ \frac{c_2}{R^2}+\frac{k_3}{R^3}=\frac{k_2}{R^2} \\ \vdots \\ \frac{c_T}{R^T}+\frac{k_{T+1}}{R^T}=\frac{k_T}{R^{T-1}} \end{cases}$$

$(c_1+k_2=Rk_1 \ ; \ \div R)$

$k_{T+1}=0$

$\Rightarrow$ /add them up/ $\Rightarrow$

$$\Rightarrow c_0+\cancel{k_1}+\frac{c_1}{R}+\cancel{\frac{k_2}{R}}+\frac{c_2}{R^2}+\cancel{\frac{k_2}{R^2}}+\dots+\frac{c_T}{R^T}+\frac{k_{T+1}}{R^T}=$$

$$=Rk_0+\cancel{k_1}+\cancel{\frac{k_2}{R}}+\cancel{\frac{k_2}{R^2}}+\dots+\frac{k_T}{R^{T-1}}$$

$$\Rightarrow \underbrace{c_0+\frac{c_1}{R}+\frac{c_2}{R^2}+\dots+\frac{c_T}{R^T}}_{\text{PV of life-time cons.}}=\underbrace{Rk_0}_{\text{PV of life-time wealth}}$$

PV of life-time cons. RV of life-time wealth

(linear technology = save at interest rate R)

Suppose also that $u(c) = \log(c)$

$\Rightarrow$ Euler Equation (EE): $c_{t+1} = \beta R c_t$

$$\Rightarrow \quad c_t = (\beta R)^t c_0$$

$$\Rightarrow \quad c_0 + \frac{\beta R}{R} c_0 + \left(\frac{\beta R}{R}\right)^2 c_0 + \dots + \left(\frac{\beta R}{R}\right)^T c_0 = R k_0$$

$$\Rightarrow \quad c_0 = R \cdot \frac{1}{1+\beta+\dots+\beta^T} k_0 = \boxed{\frac{1-\beta}{1-\beta^{T+1}} R k_0 = c_0}$$

$$\boxed{c_t = (\beta R)^t c_0}$$

$t = 1, \dots, T$

Remark: This is clearly a generalization of the consumer decision problem that you've seen in the OLG models.

This "trick" of summing up the budget constraints and cancelling intermediate k_{t+1} will not work for non-linear technology.

Then there is another approach.
Illustrate it on the linear case here, you'll have to do it for a concave $f(k)$ in the HW.

Example 2: $u(c) = \log c$, $f(k) = R k$, alternative approach

$$(EE) \Rightarrow \quad u'(c_t) = \beta R \, u'(c_{t+1}) \quad \Rightarrow \quad c_{t+1} = \beta R c_t$$

feasibility $\Rightarrow$ $c_t = R k_t - k_{t+1}$

Denote $k_{t+1} = \underbrace{s_t}_{\text{savings rate}} \cdot R k_t$

$$\Rightarrow (EE): \quad \frac{c_{t+1}}{c_t} = \frac{R k_{t+1} - s_{t+1} \cdot R \cdot k_{t+1}}{R k_t - s_t \cdot R k_t} = \beta R$$

Rearrange $\Rightarrow$ $$\frac{(1-s_{t+1}) \cdot R k_{t+1}}{(1-s_t) \cdot R k_t} = \beta R, \quad k_{t+1} = s_t \cdot R k_t$$

$$\Rightarrow \frac{(1-s_{t+1}) \cdot R \cdot \overbrace{s_t R k_t}^{k_{t+1}}}{(1-s_t) R k_t} = \beta R$$

$$\Rightarrow R \cdot \frac{s_t (1-s_{t+1})}{1-s_t} = \beta R \qquad t = 0 \dots T-1$$

Add $k_{T+1} = 0 \Rightarrow s_T = 0$

$\Rightarrow$ the optimal savings rate solve

$$\boxed{\begin{aligned} &(1-s_{t+1}) \cdot s_t = \beta (1-s_t), \quad t = 0 .. T-1 \\ &s_T = 0 \end{aligned}}$$

This is a first-order difference equation with the terminal condition $s_T = 0$ $\Rightarrow$ we can easily solve it backwards!

$$s_t = \frac{\beta}{1+\beta - s_{t+1}}$$

Remark: generally, $s_t \neq s_{t+1}$ (contrary to the infinite horizon Solow model)

$$s_T = 0 \Rightarrow s_{T-1} = \frac{\beta}{1+\beta}$$

$$s_{T-2} = \frac{\beta}{1+\beta - \frac{\beta}{1+\beta}} = \frac{\beta(1+\beta)}{1+2\beta+\beta^2-\beta} = \frac{\beta(1+\beta)}{1+\beta+\beta^2}$$

$$S_{T-2} = \frac{\beta}{1+\beta-\frac{\beta}{1+\beta}} = \frac{\quad}{1+2\beta+\beta^2-\beta} = \frac{\quad}{1+\beta+\beta^2}$$

$$S_{T-3} = \frac{\beta}{1+\beta-\frac{\beta(1+\beta)}{1+\beta+\beta^2}} = \frac{\beta\,(1+\beta+\beta^2)}{1+2\beta+\beta^2+\beta^2+\beta^3-\beta-\beta^2} = \frac{\beta(1+\beta+\beta^2)}{1+\beta+\beta^2+\beta^3}$$

- - - - show by induction that

$$S_{T-t} = \frac{\beta(1+\beta+ \dots +\beta^{t-1})}{1+\beta+ \dots +\beta^{t}} = \beta\cdot\frac{1-\beta^{t}}{1-\beta^{t+1}}, \quad t=1,\dots T$$

Rename the index $T-t \to t$ $\Rightarrow$

$$\boxed{\begin{aligned} S_t &= \beta\cdot\frac{1-\beta^{T-t}}{1-\beta^{T-t+1}}, \quad t=0,\dots,T-1 \\ S_T &= 0 \end{aligned}}$$

Still have not found $\{C_t\}_{t=0}^{T}$ or $\{k_{t+1}\}_{t=0}^{T}$ ...

$$k_{t+1} = S_t\cdot R k_t = \beta\cdot\frac{1-\beta^{T-t}}{1-\beta^{T-t+1}}\cdot R k_t$$

$$\Rightarrow C_t = R k_t - k_{t+1} = R k_t\cdot(1-S_t) =$$

$$= R k_t\cdot\frac{1-\cancel{\beta^{T-t+1}}-\beta+\cancel{\beta^{T-t+1}}}{1-\beta^{T-t+1}} = \frac{1-\beta}{1-\beta^{T-t+1}}\cdot R k_t$$

$$\Rightarrow \boxed{\begin{aligned} C_0 &= \frac{1-\beta}{1-\beta^{T+1}}\cdot R k_0 \\ C_t &= (\beta R)^{t}\cdot C_{t-1}, \quad t=1..T \end{aligned}}$$

same as in the previous example :)